

Kestrel Compound

Autonomous Investment Operating System

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How It Works (Plain English)

Layer 1: Data Sources (top of diagram)

The system reads from 14 different information feeds across the internet. Some track large crypto transactions (whales moving millions), others read crypto news sites like CoinDesk and CoinTelegraph, venture capital newsletters like a16z, market sentiment indexes, and even Telegram chat groups. Each source has a reliability score from 0.90 (most trusted) down to 0.15 (least trusted).

Layer 2: Ingestion Pipeline

When a signal arrives, the Router checks its source confidence and event importance, then routes it through four priority lanes. Dedup removes exact duplicates using a fingerprint system. A freshness watchdog makes sure nothing goes stale.

Layer 3: Storage (DuckDB)

Every signal gets stored in DuckDB - think of it as a giant, fast spreadsheet. Right now it holds 4,671 scored signals from all 14 sources. The database also tracks source accuracy, cross-source agreement, and trade history. Everything is backed up to GitHub every 4 hours.

Layer 4: Scoring Engine

Every 15 minutes, every new signal gets a score from 0.0 to 0.384. The score considers: how trustworthy the source is, how fresh the event is, whether multiple sources agree, and whether it's about a major asset like BTC or ETH. Only signals above 0.20 get traded on. Currently 148 signals pass this bar.

Layer 5: Trading Execution

Top signals flow to our trading bots. FreqTrade handles cryptocurrency trading on Coinbase (BTC, ETH, SOL). MMR handles stock trading on Interactive Brokers. Both are currently running on simulated money (paper trading) to prove the strategy works before real money goes in. A supervisor manages both bots with a shared \$50/day budget guard.

Layer 6: Monitoring

The system runs itself on automated timers. A watchdog checks health every minute. A scoring job runs every 15 minutes. RSS feeds scrape every 4 hours. A Perplexity research query generator runs every 30 minutes. The whole system is designed to run 24/7 with zero manual intervention.

The Money Flow

1. Signals come in from 14 sources
2. Each signal gets scored for quality
3. High-quality signals (score > 0.20) enter the ranked queue
4. Trading bots read the queue and execute buy/sell orders
5. Results get logged back to the database
6. The system learns from past trades to improve future scoring

Current Status

All systems are running on simulated (paper) money. No real funds are at risk. The next steps are: tune the signal detector to reduce noise, enable boot persistence for reliability, integrate CryptoQuant (our highest-value remaining source), and deploy the visual dashboard. Once the strategy proves itself over time, we flip to real money.